

SmartCare v0.3 - Domain Model Workbook

Week 6 student resource

Requirement-to-Concept Trace

Requirement	Concept	State/Behaviour	Decision
FR-01: Create patient	Patient	State: name, contact info	Accepted
FR-02: Search for patient	None (System action)	findPatient()	Left out of domain model to avoid creating a manager class
FR-03: Create appointment	Appointment	State: date, time, status	Accepted
FR-04: Stop double bookings	Appointment	Behaviour: check availability	Accepted
FR-05: Update appointment status	Appointment	State: status (like cancelled)	Accepted
FR-06: Show appointment history	Patient	Behaviour: get history list	Accepted
FR-07: Create practitioner	Practitioner	State: name, ID	Accepted
FR-08: View daily schedule	Practitioner	Behaviour: view schedule	Accepted

CRC Cards

Patient

Responsibilities	Collaborators
Knows past appointment history	Appointment
Knows personal details (ID, Full Name, Phone number, etc.)	Appointment

Practitioner

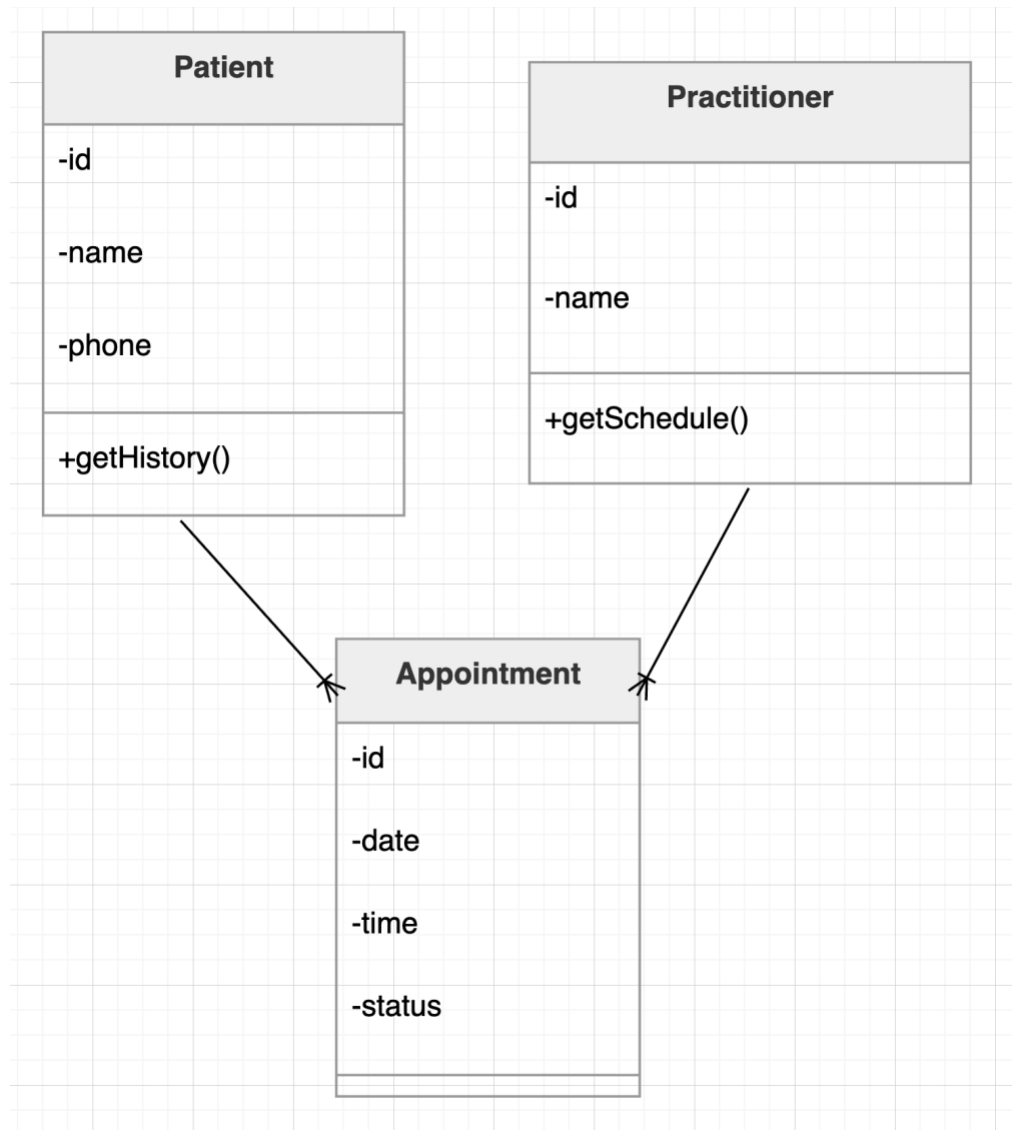
Responsibilities	Collaborators
Know practitioner details	Appointment
Know assigned daily schedule	Appointment

Appointment

Responsibilities	Collaborators
Know scheduled date, time and status	Appointment
Link a specific patient to a practitioner	Patient and practitioner

UML Class Diagram

Insert/draw UML here. Include defensible relationships and multiplicities.



Design Rationale

Explain class selection, responsibility allocation and key relationships.

I kept the design super simple with just three main classes, Patient, Practitioner, and Appointment. The management specifically asked for a “small, maintainable” system, so adding extra classes like Clinic or Billing would just overcomplicate things. Instead of a massive central system, patients track their own history, and practitioners track their own schedules.

AI Design Review Record

AI suggestion	Evidence	Decision	Reason	Model change
Add a NotificationManager class	None	Rejected	The brief never mentioned SMS or email reminders.	None
Add a Status field to Appointment	FR-05	Accepted	We need a way to show if an appointment is Cancelled vs Scheduled.	Added “status” to Appointment
Add a Billing class	None	Rejected	Client only asked for patient, practitioner and appointment tracking.	None